

Leaching of Estrogenic Hormones from Manure-Treated Structured Soils

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The threat to the aquatic environment posed by root zone leaching of estrogens from manure-treated fields has hitherto been overlooked. The steroid hormones 17 β -estradiol (E2) and its degradation product estrone (E1) are of particular environmental concern as both are abundant in slurry from pregnant and cycling pigs and both are potential endocrine disruptors (lowest observable effect level (LOEL) 14 and 3.3 ng/L, respectively). The present one-year study examines the transport of E1 and E2 from manure to tile drainage systems at two field sites on structured, loamy soil. The estrogens leached from the root zone to tile drainage water in concentrations exceeding the LOEL for as long as 3 months after application, with the maximum recorded concentration of E1 and E2 being 68.1 and 2.5 ng/L, respectively. Transport of estrogens from the soil to the aquatic environment was governed by pronounced macropore flow and consequent rapid movement of the estrogens to the tile drains. These findings suggest that the application of manure to structured soils poses a potential contamination risk to the aquatic environment with estrogen, particularly when manure is applied to areas where the majority of streamwater derives from drainage water.

Introduction

While the contamination threat posed by nutrient leaching from manure-treated fields is well recognized, that posed by leaching of steroid hormones from the manure has received much less attention. The estrogens 17 β -estradiol (E2) and its degradation product estrone (E1) are of particular environmental concern due to their marked endocrine disrupting properties. Laboratory studies suggest that the estrogenic effects of these compounds on fish species occur at concentrations of a few nanograms per liter. In a recent review (1) the predicted-no-effect-concentration (PNEC) was 1.0

ng E2/L and 3–5 ng E1/L, while the lowest observable effect level (LOEL) affecting vitellogenin production in fish species (juvenile female rainbow trout) was found to be as low as 3.3 ng E1/L and 14 ng E2/L (2).

Several studies (3–7) and a review (8) of surface runoff from agricultural fields suggest that estrogens in manure could potentially contaminate the aquatic environment. While surface runoff is thus recognized as an important pathway for the transport of estrogens to the aquatic environment, little is known about the pathway reported here, namely leaching of these substances through the soil to either shallow groundwater or (via drainage water) to surface water. The risk that E1 and E2 will leach is generally considered to be low given their high sorption (9–14) and/or fast dissipation (9, 11–17) in well-aerated topsoil. Anoxic conditions retard their degradation, however (18). For more detail see Section 1 of the Supporting Information.

Most assessments of the risk of estrogens leaching from soil do not take into account the potential impact of preferential transport. Thus either the process is not included at all in batch studies (9, 10, 13, 15–18), or the studies involve systems in which the flow regimes are not subject to macropore flow such as repacked soil columns (11, 12, 14). In structured soils, however, preferential flow can have a major impact on leaching. Field studies on structured soils thus indicate considerable preferential transport of xenobiotics such as pesticides (19–21). Moreover, studies of the E2 concentration in cave streams fed by karstic aquifers suggest rapid preferential transport of E2 through karst systems (22–24).

The present study is the first controlled field study of edge-of-field leaching of estrogens from manure-treated structured soils. Our aim was to assess the risk that these potential contaminants leach into the aquatic environment when manure is applied to fields in accordance with current regulations.

Experimental Section

Pig (*Sus scrofa domesticus*) slurry from a farm, having sows in all stages of their reproductive cycle as well as their offspring to a weight of 30 kg (weaner production) was applied to two tile-drained loamy field sites (Estrup and Silstrup) in accordance with Danish regulations (25) concerning manure doses and method of application. The concentrations of E1 and E2 were determined on tile drainage water samples collected during typical storm events over a 12-month period following slurry application.

Site Description. The two systematically tile-drained loamy field sites were located at Silstrup and Estrup (Figure S1 in Supporting Information) on glacial till, a widespread geological formation covering about 43% of all farmland in Denmark (26, 27). At both sites the tile drains were at an average depth of about 1 m. Spacing between laterals at the Silstrup field was approximately 17–18 m whereas the distance at Estrup varied between 12 and 15 m. The water table at each site is relatively shallow, located 1–3 m bgs. At both field sites the uppermost meter of the soil is heavily fractured and bioturbated, containing 100–1000 biopores/m² (28).

The Silstrup field site covers 1.71 ha and slopes gently 1–2°. The site is located on a glacial moraine of Late Weichselian Age and has been exposed to weathering, erosion, leaching, and other geomorphologic processes for about 16,000 years (29). There is little variation between the site's two pedological profiles, one of which is classified as Alfic Argiudoll and the other as Typic Hapludoll. The latter

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TABLE 1. Slurry Application Method, Dose, Percentage Dry Matter, Wet Volume, Measured Slurry Concentration of E1 and E2 and Total Amount of E1 and E2 Applied

site	application method ^a	dose (t/ha)	dry matter (%)	wet volume ^b (mm)	E2 ^c (μg/kg DM)	E1 ^c (μg/kg DM)
Estrup						
04/18/2005	injected (8–10 cm) and plowed down (20 cm)	53.0	3.15	5.1	< 2.5	866 (1.4)
Silstrup						
04/23/2005	injected (8–10 cm)	30.2	8.3	2.8	392 (1.0)	643 (1.6)
08/29/2005	trailer-hosed and plowed down (20 cm)	15.8	2.9	1.5	< 2.5	1.068 (0.5)

^a Depth of tillage is indicated in parentheses. ^b Bulk density of slurry estimated to be 1.0 g/cm³ (Jens Petersen, personnel communication).
^c Number in parentheses indicates the total amount of E1 and E2 applied (g/ha).

is included in Table S1 in Supporting Information. The geological settings consist of rather homogeneous clay till rich in chalk and chert with local occurrences of thin silt and sand beds (28).

The Estrup field site covers a cultivated area of 1.26 ha and is virtually flat. Compared to the Silstrup site, the Estrup site is highly heterogeneous with considerable variation in both topsoil and aquifer characteristics. Such heterogeneity is quite common for this geological formation, however. The site is located on a glacial moraine of Saalian Age and has been exposed to weathering, erosion, leaching and other geomorphologic processes for about 105,000 years, which is about 90,000 years longer than the Silstrup site (29). The geological structure is complex, comprising a clay till core with deposits of different age and composition. Of the three pedological profiles at the site, one is classified as Aquic

Argiudoll, one as Abruptic Argiudoll, and one as Fragiaquic Glossudalf, the latter two being shown in Table S1.

Agricultural Management. The Silstrup and Estrup field sites were both cultivated conventionally for the area as regards crop rotation, fertilization, and soil tillage. Manure from a weaner holding was applied by injection prior to sowing of spring barley at both the Estrup and Silstrup sites and additionally by trailer hosing in autumn prior to sowing of winter rape at the Silstrup site. The slurry was applied in accordance with Danish regulations concerning dose and application method, as given in Table 1. Prior to application and sampling, the slurry tank was homogenized for 1 h, using a slurry agitator (Kimidan Multimixer, Denmark).

Finally, it should be noted that the Silstrup site was originally part of a dairy holding. The most recent applications of cattle slurry prior to the present study were 36.5 t/ha in

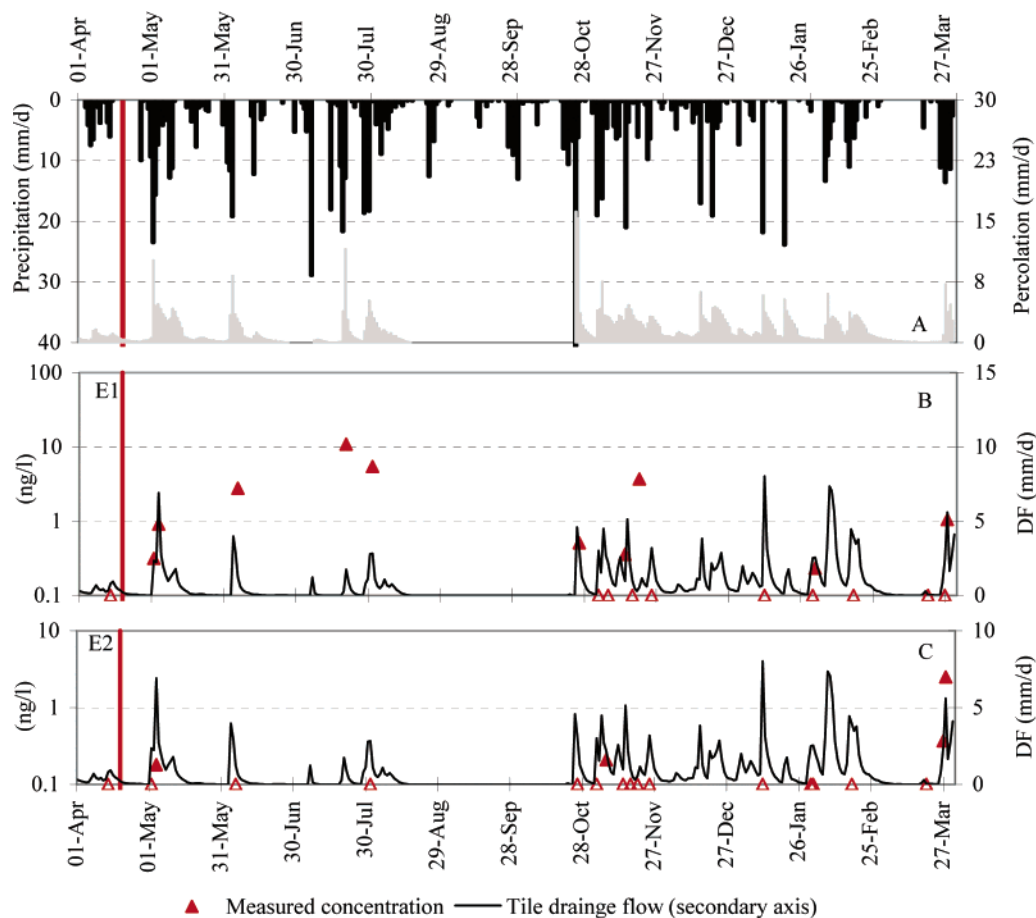


FIGURE 1. Precipitation (black hanging bar on primary axis) and simulated percolation 0.6 m below ground surface (gray columns on secondary axis) at the Estrup field site (A) together with the concentration of E1 (B) and E2 (C) in the tile drainage flow (DF on secondary axis). Application of the slurry (injected and plowed down) is indicated by a red vertical bar. Open triangles indicate measured concentrations below the limit of detection (0.1 ng/L).

TABLE 2. Annual Water Balances (04/01/05–3/31/06), Accumulated Precipitation (First Precip.) and Percolation (First Percol.) Occurring within the First Month after Application of Manure, and Measured Concentrations of Estrogens E1 and E2 in Drainage Runoff

	normal precip. ^a (mm)	prec. ^b (mm)	E _{act} ^c (mm)	Dr. ^d (mm)	GwR ^e (mm)	application date	first precip. (mm)	first percol. ^c (mm)	n ^f	E1		E2	
										det. ^g (%)	C _{max} ^h (ng/L)	det. (%)	C _{max} (ng/L)
Estrup	997	922	425	263	234	4/18	107	57	20	55	10.9	25	2.5
Silstrup	1006	785	447	79	259	4/23 & 8/29	57 & 35	6 & 0	7	57	68.1	14	1.8

^a Normal annual precipitation (normal precip.) based on time series for 1961–1990 is included for comparison. ^b Precipitation (prec.) corrected to the soil surface according to the method described in (33). ^c Actual evapotranspiration (E_{act}) and percolation (percol.) are estimated using the MACRO model, version 5.1 in (34). ^d Dr.: Measured drainage runoff ^e GwR.: Groundwater recharge calculated as precip. – E_{act} – Dr. ^f n: number of analyzed samples. ^g Det.: Detection rate (% of samples containing E1 and E2 with limit of detection being 0.1 ng/L). ^h C_{max}: maximum concentration detected

2000 and 40.3 t/ha in 2002. The Estrup site was formerly part of a crop and mink farm, with manure of various types therefore having been applied in the past. The most recent application of slurry prior to the present study was 60.8 t/ha of cattle slurry in 2003.

Monitoring. The concentrations of E1 and E2 were determined on drainage water samples collected during typical storm flow events at the two field sites over a 12-month period following slurry application. Following the onset of the storm events, drainage water was sampled flow-proportionally for approximately 1 day. In order to obtain the weighted average concentration for each storm event, the chemical analysis was performed on pooled water samples containing all the subsamples collected during the storm event. The “typical” storm events were defined as events causing the water level and accumulated flow rate within the preceding 12-hour period to exceed predefined levels that depended on the month of the year. The predefined water level rise and accumulated flow rate were set/adjusted individually for each site based on experience (30). An electronic warning system activated at the onset of each storm event enabled the drainage water to be collected and conserved (see Analytical Methods) shortly thereafter, with the storage time in the cooled ISCO 6700 sampler (Teledyne Isco, Inc, US) being about 1 day. Prior to monitoring, a drainage water sample was collected and analyzed for E1 and E2. As storm events did not occur at this time, the samples were collected as manual grab samples. Likewise a single sample was collected as a manual grab sample at Silstrup on May 23, 2005. Precipitation was measured at each site using a tipping bucket rain gauge system.

To avoid unintended leaching of contaminants due to the installation and presence of sampling equipment in the ground, all installation work and soil sampling deeper than 20 cm were restricted to a buffer zone bordering the slurry-treated area. The drainage system at the Silstrup site was established in the 1960s, while that of the Estrup site was established prior to 1965; both are at a depth of 1 m. The systems were modified prior to monitoring by cutting off and blocking the drainpipes entering from upstream fields to ensure that tile-drainage water was only collected from the treated area; all modifications being made outside the field. Water coming from upstream fields was then diverted around the test field through a corrugated and perforated PVC pipe in an envelope of filter sand. The drainage water of the test fields was directed into newly established wells fitted with Thomson weirs (30° V-notch). The water level behind the weirs was measured automatically using a pressure transducer (Druck, PDCR1830, UK) coupled to a CR10X data logger (Campbell Sci., UK). Drainage water was sampled flow-proportionally using two cooled ISCO 6700 samplers. Sampling frequency was set by experience, as described elsewhere (30), with 200-mL subsamples being collected for every 3000 L of drainage flow during the winter

season (September–May) and for every 1500 L during the summer period (June–August).

Sample Preparation and Analytical Methods. *Water Samples.* Water samples (1000 mL) were immediately adjusted to pH 3 and amended with deuterated internal standards (*d*₄-estone and *d*₅-17β-estradiol); storage time prior to this conservation being about 1 day (see Monitoring). Until required for analysis the water samples were stored at –18 °C. Thereafter the samples were filtered through glass fiber and applied onto solid-phase extraction cartridges (SPE). The SPE cartridges were air-dried for 1 h and if necessary stored at –18 °C. After elution of the SPE cartridges with 5 mL of acetone the eluate was purified (silica gel), derivatized, and reconstituted in heptane before injection in the GC–MS/MS system. Quantification and detection of the analytes were accomplished using a gas chromatographic–tandem mass spectrometry system (GC–MS/MS) consisting of a gas chromatograph (Varian CP-3800) with a programmable temperature vaporizer injector and a triple stage quadrupole mass spectrometer (Varian MS 1200) operated in the electron impact mode. The analytical GC–MS/MS methods followed the procedure described in refs 31 and 32.

Manure Samples. The manure samples were frozen at –18 °C, freeze-dried to complete dryness, and stored at –18 °C until needed for analysis. The freeze-dried manure was mechanically homogenized and extracted by pressurized liquid extraction (PLE) using methanol/acetone (1:1). The PLE extract was evaporated and reconstituted in 5 mL of acetonitrile/MilliQ-water (2:3) and centrifuged at 900g for 3 min. The supernatant was transferred to a bottle, while another 5 mL of acetonitrile/MilliQ-water (2:3, v/v) was added to the precipitate. The sample was mixed and centrifuged, and the supernatant was again transferred to the same blue-cap bottle. The two extracts were combined, diluted with 150 mL MilliQ-water, and adjusted to pH 3.

The diluted extracts were solid-phase extracted as described for water samples. The SPE extracts were spiked with 5-fold the normal volume of internal standards and thereafter divided into five subsamples for addition of matrix-matched standard. The subsamples were then purified using silica gel and derivatized prior to GC–MS/MS analysis as described for water samples.

Quality Assurance. Deuterated internal standards enabled correction for loss of estrogens during sample preparation. A derivatization standard (17-acetate-17β-estradiol) was used to monitor the quality of the derivatization step. A co-analyzed instrument control standard (3-methyl estone) was used to monitor the quality of the GC–MS/MS system. Further details on quality assurance including control charts are given in the Supporting Information (Section 4).

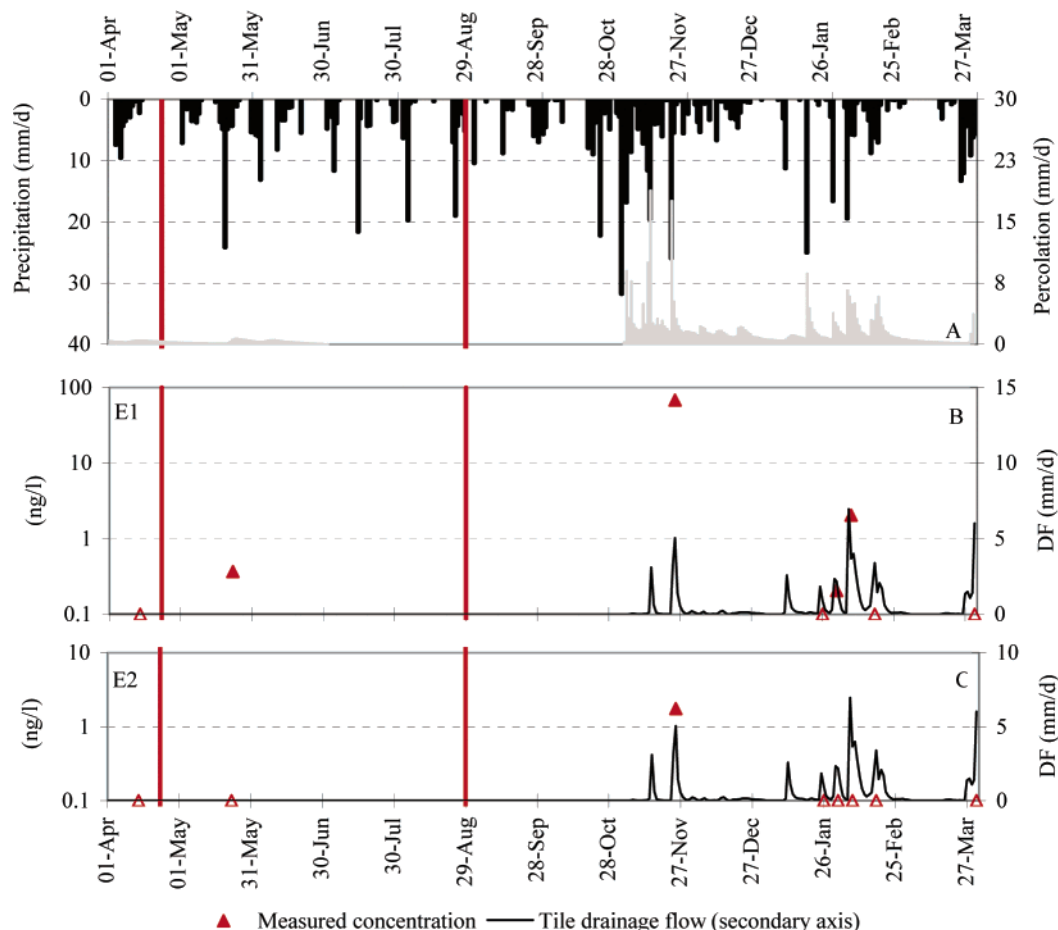


FIGURE 2. Precipitation (black hanging bar on primary axis) and simulated percolation 1 m below ground surface (gray columns on secondary axis) at the Silstrup field site (A) together with the concentration of E1 (B) and E2 (C) in the tile drainage flow (DF on secondary axis). Application of the slurry (injected in April and trailer hosed/plowed down in August) is indicated by red vertical bars. Open triangles indicate measured concentrations below the limit of detection (0.1 ng/L).

Results

Estrogens E1 and E2 were found to leach from the root zone into the tile drains at both field sites. A few samples (2 of 20 at Estrup and 1 of 7 at Silstrup) had concentrations of E1 exceeding LOEL (the maximum recorded concentrations being 68.1 ng/L), and about half of the analyzed samples had detectable concentrations of E1.

At the Estrup site, no precipitation fell during the first 9 days after application of the slurry (Figure 1). Drainage runoff responded rapidly to the first storm event. Heavy storm events (23 and 16 mm/d) on May 2 and 3 induced rapid leaching of both E1 and E2 in concentrations as high as 0.9 and 0.2 ng/L, respectively. Thereafter E2 only leached on a few occasions, each time in a low concentration that never exceeded the LOEL. In contrast, many of the subsequent flow events induced leaching of E1, and the two samples with concentrations exceeding the LOEL were collected 3 months after application of the slurry. The maximum concentration recorded was approximately 11 ng E1/L and E1 was detected as long as 11 months after application of the slurry, although detections following the long dry summer months were sporadic, being limited to 5 of 14 samples (Figure 1).

At the Silstrup site, precipitation, percolation, and corresponding drainage runoff were much lower than at the Estrup site and fewer samples were therefore obtained, particularly during the first month after application of the slurry. The situation at Silstrup may thus be viewed as a “best case” scenario (Table 2 and Figure 2). Nevertheless,

the heavy storm event on November 24 (26 mm/d) induced leaching of E1 at a concentration of 68.1 ng/L 3 months after the second applications of slurry (Figure 2). Subsequent precipitation events involved less than 25 mm/d and did not cause major leaching of E1. E2 only leached on one occasion, with the concentration being low (1.8 ng/L) and never exceeding the LOEL.

When evaluating the leaching risk it should be noted that precipitation conditions during the monitoring period were not particularly wet, with precipitation at the Estrup and Silstrup sites being 8% and 23% lower than normal, respectively (see Table 2). The precipitation pattern, in terms of number and seasonal distribution of heavy storm events exceeding 20 mm/d that caused leaching of estrogens in concentrations exceeding the LOEL, is not unusual for the regions where the two sites are located (based on 16-year time series, data not shown).

Discussion

Our findings demonstrate that estrogens are able to leach from manure-treated soil and that E1 in particular can leach in high concentrations 3 months after application of the slurry. This finding contrasts with a number of laboratory studies suggesting that the risk that estrogens will leach from manure-treated fields is low (11–13, 15, 16). The apparent inconsistency may be due to the fact that our field study results, unlike those studies, are influenced not only by the macroporous transport but also whatever possible effects may stem from a lower soil temperature,

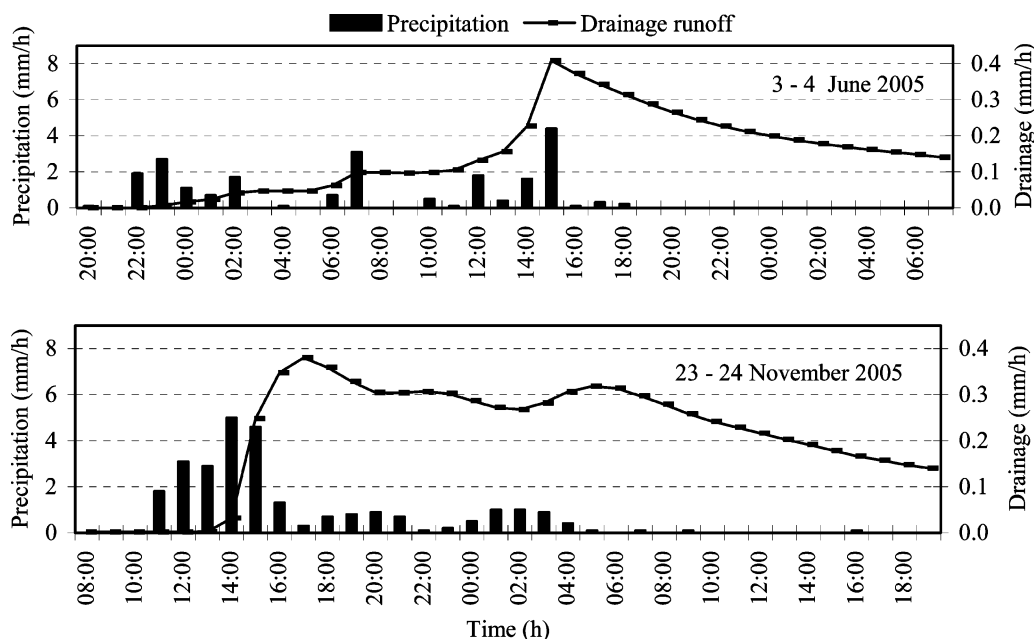


FIGURE 3. Hourly precipitation, drainage runoff, and estimated percolation for two storm events at the Estrup (upper) and Silstrup (lower) field sites.

the manure itself, as well those exerted by the tillage operations.

The lack of macropores in such laboratory setups (11–13, 15, 16) provides better conditions for estrogen sorption and/or degradation due to the longer residence time and the better contact between the infiltrating water and the surrounding soil matrix. The leaching risk as assessed in such laboratory systems will thus be much lower than under field conditions where macropores are present. At both the Estrup and Silstrup sites, leaching appears to be influenced by preferential transport as evidenced by the soil hydraulic properties (21, 35), fast solute transport, and drainage water dynamics. The drainage water was very sensitive to precipitation input, responding within a few hours to even small precipitation events (Figure 3). The rapid response occurred when the overlying soil was not close to saturation (water saturation of the upper 25 cm of the soil, being 82% at Estrup and 72–84% at Silstrup) indicating that the flow regime was influenced by preferential flow allowing parts of the water to bypass the soil matrix, thereby allowing estrogens to leach rapidly to the drainage water. Similar rapid transport of estrogen was also observed with first detection of estrogen occurring only 14 days and 30 days after application at Estrup and Silstrup, respectively. The observed transport time was thus considerably faster than that occurring through the soil matrix. Piston transport through the low permeability soil matrix (Table S1 in Supporting Information) would involve a travel time to the drainage system of Estrup and Silstrup of about 98 and 69 days, respectively. This finding is consistent with previous transport studies conducted at the Estrup and Silstrup sites (21, 35), as well as with other field studies demonstrating rapid macropore-mediated transport of xenobiotics such as pesticides (for a review see (19, 20)).

That estrogen E1 leached several months after application of the slurry when drainage runoff resumed after the dry summer/autumn months (Figures 1B and 2B) conflicts with previous studies suggesting rapid dissipation of the E1 generated from E2 (11, 12, 14, 16).

The latter laboratory studies were conducted either at ambient temperature (11, 12, 14, 16) or at 30 °C (16). In our field study the average temperature of the plow layer ranged between 7 °C in April to 17–18 °C in July (data not shown). In the loamy soils studied by Colucci et al. (15) soil

temperature was found to influence the total mineralization of E2. Mineralization (61 days) decreased with decreasing temperature, with measured values being 14.7%, 14.1%, 10.9%, 6.0%, and 3.6% at 37, 30, 19, 10, and 4 °C, respectively. Low soil temperatures at the time of manure application in April (Estrup, Silstrup) might have retarded E1 degradation in our study. However, E1 also persisted at Silstrup, even though the soil temperatures during summer months (June, July, and August) were between 14 and 17 °C.

Moreover, the previous studies (11, 12, 14, 16) were all conducted under aerobic conditions, without the addition of manure, however, and hence do not take into account possible combined effects of the manure matrix and manure application/soil tillage methods on leaching of estrogens. No experimental data are available regarding the importance of these factors in combination, although it would be expected that the higher organic content of the manure would result in higher sorption of estrogens.

When estrogen-containing manure is applied to and incorporated into the soil, conditions in the manure are likely to remain anoxic for quite some time, thereby retarding degradation of the estrogens. Evidence that conditions in manure-treated soil are anoxic is provided by studies in which a slurry string was constructed by “sandwiching” 16 mm of manure between 35 mm of soil in packed columns (36). The anoxic slurry string was found to re-aerate very slowly, with the oxygen penetration depth being only 1 mm and 2 mm after 3 days and 3 weeks, respectively. At the Silstrup site the first slurry was applied by injection. As the soil was not subsequently tilled, the string of slurry would have remained intact in the soil. With the second application of slurry at the Silstrup site and the single application at the Estrup site the slurry strings were disrupted by plowing, thereby allowing more rapid re-aeration of the slurry. Nevertheless, compared with the above-mentioned studies of the fate of estrogens under aerobic conditions, estrogen incorporated in a slurry string is likely to remain under anoxic conditions for a much longer period. Redox conditions (aerobic vs anoxic) are known to significantly affect degradation, as evidenced by batch experiments in agricultural top soil, river bed material, and aquifer material (9, 18, 37). See Table S2 in the Supporting Information for details. Experimental data on the redox condition at our field sites however are not available, why

a firm conclusion on whether the long persistence of E1 is influenced by anaerobic conditions in the slurry retarding degradation of the estrogens cannot be drawn.

Finally, the fact that E2 leached in lower concentrations than E1 and for a much shorter period of time following slurry application is probably attributable to (1) the applied doses of E2 being lower than that of E1 (Table 2) and (2) E2 being degraded to E1. The latter is in line with several laboratory studies of the fate of E2 (9, 11–17) and a monitoring study (6) suggesting that E2 rapidly dissipates through degradation to E1. That measured concentration of E2 was lower than E1 in all manure applied (Table 2) indicates that parts of this degradation occur already during storage of manure.

As far as we are aware this is the first quantitative report on edge-of-field leaching of estrogens from manure-treated structured soil. Similar rapid preferential transport of E2 has been reported for karst systems (22–24) where the concentration of E2 in cave streams fed by karstic aquifers lay within the range 3–80 ng/L (22), 6–66 ng/L (23), and 30–35 ng/L (24). In these systems the appearance of E2 in the streamwater was attributable to the direct rapid hydrological connection between the surface water and the groundwater due to which the residence time of the streamwater was only 4 h–4 days (24) and 6–8 days (23).

Implications. Feminization of male fish due to estrogen contamination of the aquatic environment has frequently been reported in Denmark and elsewhere over the past 10–15 years (38, 39). The estrogen contamination is primarily suspected to originate from wastewater treatment plants and to some extent also from surface runoff from manure-treated fields. Our finding that artificially drained well-structured soils treated with manure may pose an additional contamination risk is of broad environmental importance since it implies that the area of agricultural land from which contaminants such as estrogens may potentially be transported to the aquatic environment is much larger than previously believed. This is due to the fact that intensively farmed land is commonly drained systematically in many countries, e.g., England and Wales (40) and Denmark. In Denmark, for example, 62% of all agricultural land is tile-drained. Considering just the macropore-rich loamy soils, the percentage is as high as 89% (41). The risk of contamination is further heightened by the fact that a substantial proportion of streamwater in Denmark derives from drainage water. In a typical Danish moraine clay catchment – the Suså river basin – 60% of the streamwater derives from drainage water, whereas the proportion deriving from surface runoff is negligible (42, 43). This highlights the much greater potential environmental importance of leaching as a pathway for estrogen transport to the aquatic environment from manure-treated fields than surface runoff.

This newly identified pathway whereby estrogen applied to fields in manure may reach the aquatic environment is also highly relevant in relation to the increasing intensification of livestock production in many countries. The present trend in pig production is toward fewer but larger and more specialized facilities producing either weaners or finishing pigs (44, 45). Estrogen concentrations are reported to be higher in slurry from weaners than from finishing pigs (46) as hormone excretion is higher in pregnant or cycling animals (8, 47). Specialization of weaner production and the consequent increasing concentration of sows on fewer holdings thus enhances the estrogen content of the slurry from such holdings and consequently the risk that estrogen might leach from fields treated with the slurry.

Our findings indicate an urgent need for further research into the risk of estrogen contamination of the aquatic environment by leaching from manure-treated fields. Knowledge of the combined effect of the manure matrix and the

manure application/soil tillage methods on estrogen persistence and leaching is particularly lacking.

Acknowledgments

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Supporting Information Available

Material supporting the Introduction, figures showing the location of the field sites, physical and chemical properties of selected soil profiles, quality assurance data for chemical analysis, and literature reports of laboratory batch studies comparing aerobic and anaerobic degradation. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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